

# Module - 01

**Introduction to Computer Networks and Network Layer**

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# 1.1 Networks

**A network is the interconnection of a set of devices capable of communication.** In this definition, a device can be a host (or an end system as it is sometimes called) such as a large computer, desktop, laptop, workstation, cellular phone, or security system.

## 1.1.1. Network Criteria

- A network must be able to meet a certain number of criteria. The most important of these are **performance**, **reliability**, and **security**.

### **Performance**

- Performance can be measured in many ways, including **transit time** and **response time**.

### **Reliability**

- In addition to accuracy of delivery, network reliability is measured by the **frequency** of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.

### **Security**

- Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

## 1.1.2. Physical Structures

- **Type of Connection**

- A network is two or more devices connected through links.
- There are two possible types of connections: **point-to-point** and **multipoint**.

### Point to Point

- A point-to-point connection provides a dedicated link between two devices.

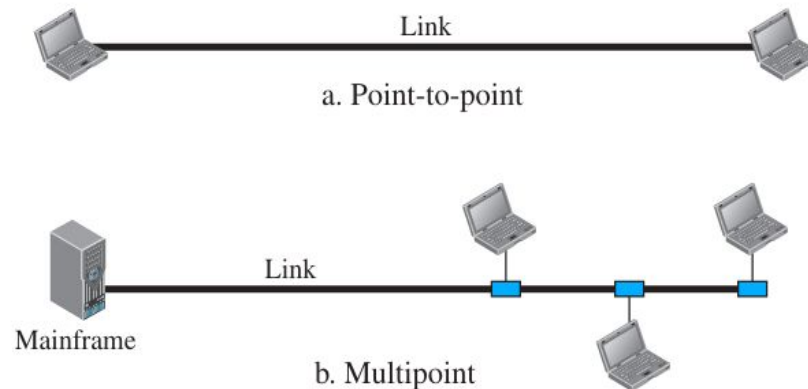
### Multipoint

- A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link (see Figure 1.3b).

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**Figure 1.3** *Types of connections: point-to-point and multipoint*

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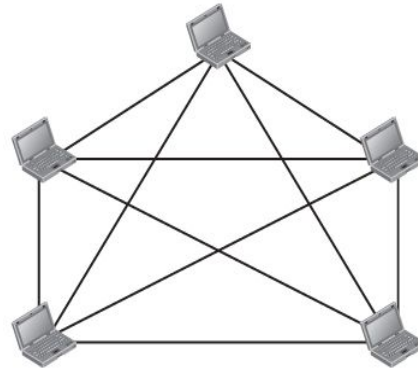
- **Physical Topology**
- The term physical topology refers to the way in which a network is laid out physically.
- **Mesh Topology**
  - In a mesh topology, every device has a dedicated point-to-point link to every other device.
  - To find the number of physical links in a fully connected mesh network with  $n$  nodes, we first consider that each node must be connected to every other node. Node 1 must be connected to  $n - 1$  nodes, node 2 must be connected to  $n - 1$  nodes, and finally node  $n$  must be connected to  $n - 1$  nodes. We need  $n(n - 1)$  physical links.
  - However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2.
  - In other words, we can say that in a mesh topology, we need  $n(n - 1) / 2$  duplex-mode links.

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**Figure 1.4** *A fully connected mesh topology (five devices)*

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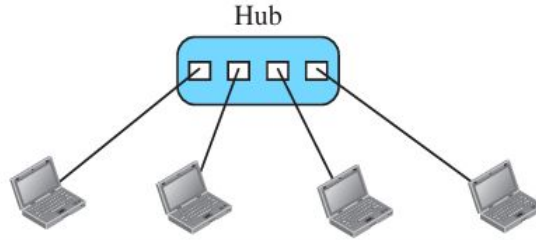
$n = 5$   
10 links.



- **Star Topology**

- In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a **hub**.

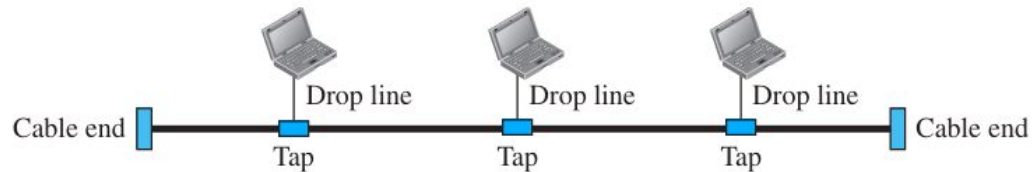
**Figure 1.5** *A star topology connecting four stations*



- **Bus Topology**

- The preceding examples all describe point-to-point connections. A bus topology, on the other hand, is multipoint. One long cable acts as a backbone to link all the devices in a network.

**Figure 1.6** *A bus topology connecting three stations*



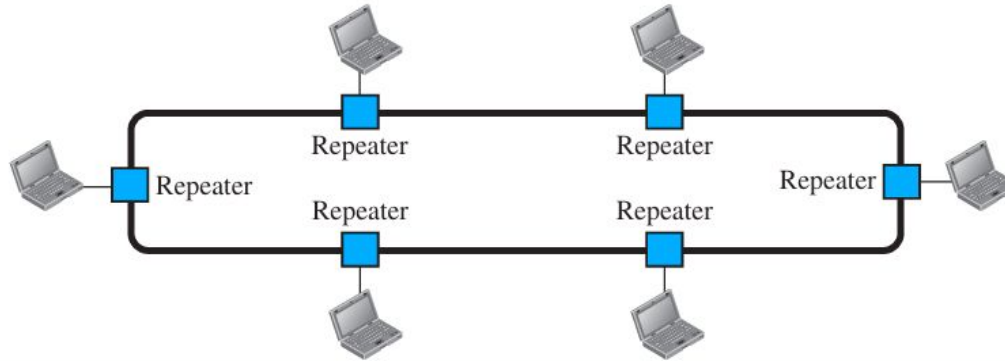
- **Ring Topology**

- In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.

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**Figure 1.7** *A ring topology connecting six stations*

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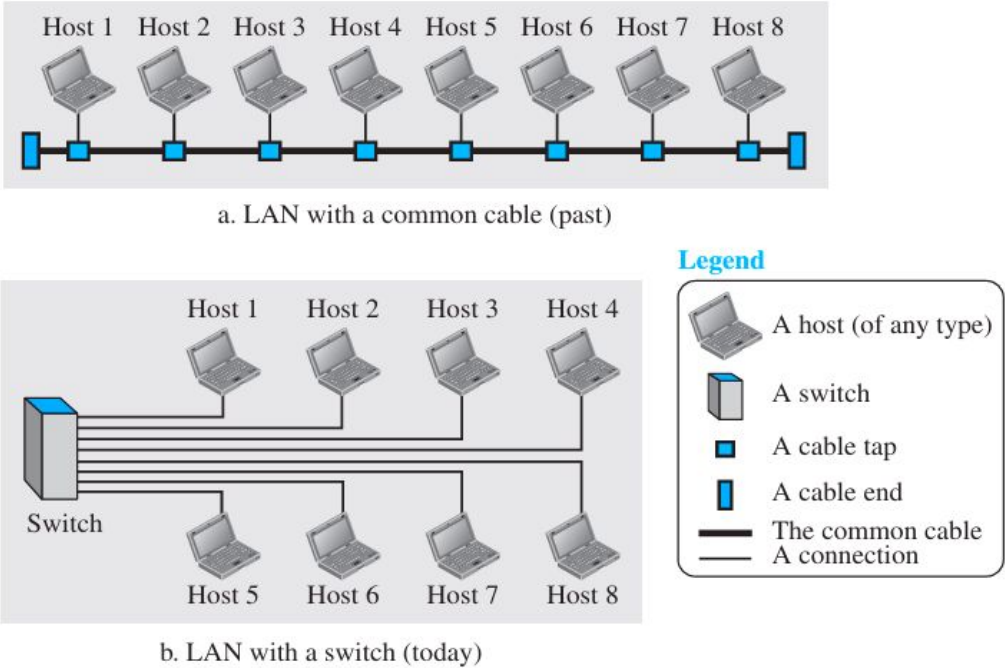
## 1.2. Networks Types

- We use a few criteria such as size, geographical coverage, and ownership to make this distinction. After discussing two types of networks, LANs and WANs, we define switching, which is used to connect networks to form an internetwork (a network of networks).

# Local Area Network

- A local area network (LAN) is usually privately owned and connects some hosts in a single office, building, or campus.

**Figure 1.8** *An isolated LAN in the past and today*





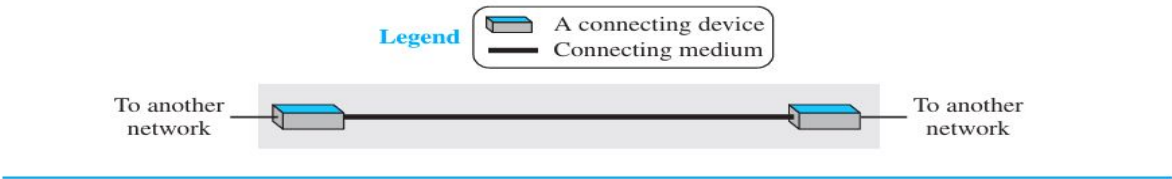
# Wide area Network

- A wide area network (WAN) is also an interconnection of devices capable of communication. However, there are some differences between a LAN and a WAN.

## Point to Point WAN

- A point-to-point WAN is a network that connects two communicating devices through a transmission media (cable or air).

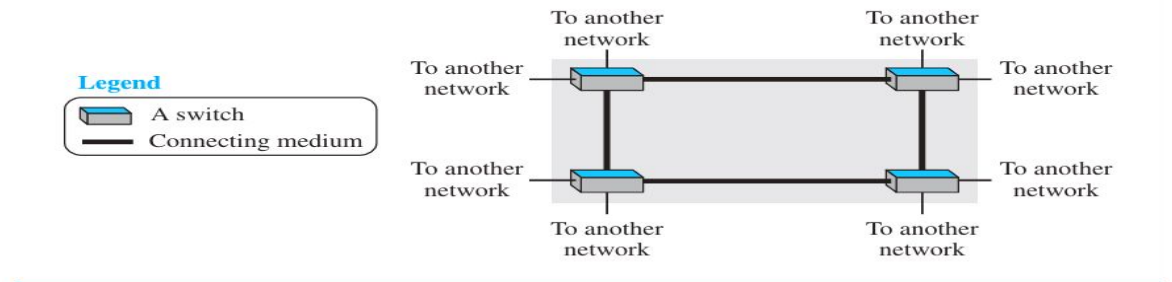
**Figure 1.9** A point-to-point WAN



## Switched WAN

- A switched WAN is a network with more than two ends.

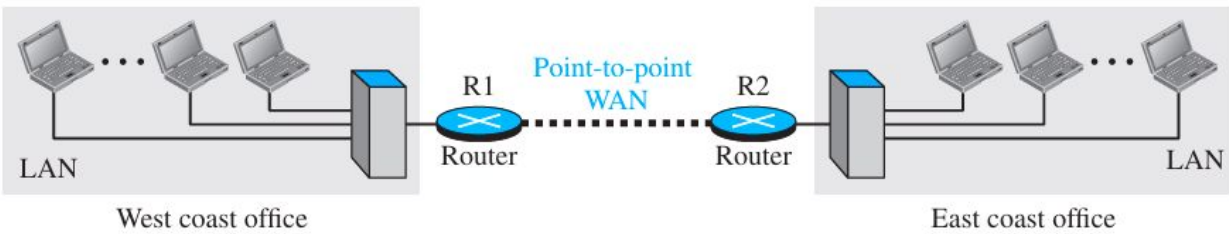
**Figure 1.10** A switched WAN



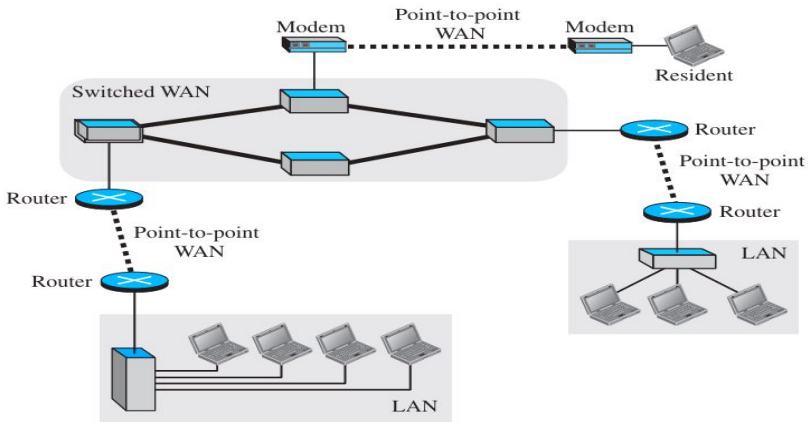
# Internetwork

- Today, it is very rare to see a LAN or a WAN in isolation; they are connected to one another.
- When two or more networks are connected, they make an **internetwork**, or **internet**.

**Figure 1.11** *An internetwork made of two LANs and one point-to-point WAN*



**Figure 1.12** *A heterogeneous network made of four WANs and three LANs*



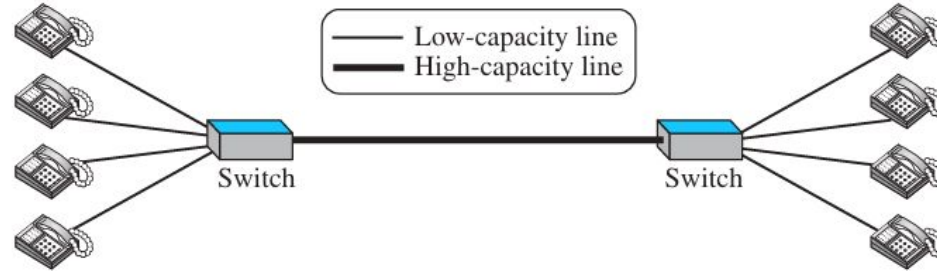
### 1.2.3. Switching

- An internet is a switched network in which a switch connects at least two links together. A switch needs to forward data from a network to another network when required. The two most common types of switched networks are **circuit-switched** and **packet-switched** networks.

#### Circuit switching Network

- In a circuit-switched network, a dedicated connection, called a circuit, is always available between the two end systems; the switch can only make it active or inactive.

**Figure 1.13** *A circuit-switched network*



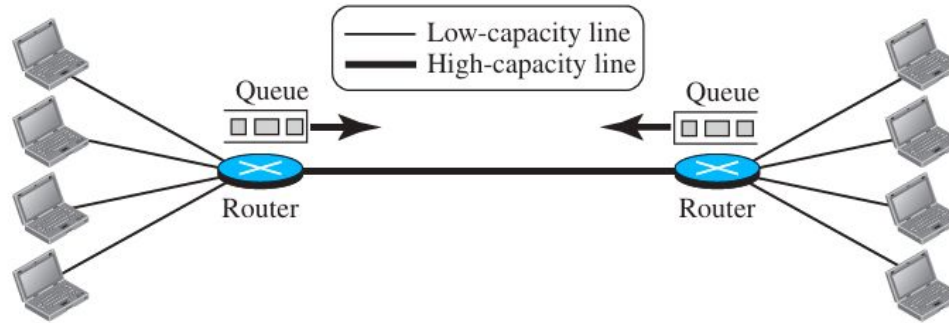
## Packet Switched Network

- In a computer network, the communication between the two ends is done in blocks of data called **packets**.
- This allows us to make the switches function for both **storing and forwarding** because a packet is an independent entity that can be stored and sent later.

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**Figure 1.14** *A packet-switched network*

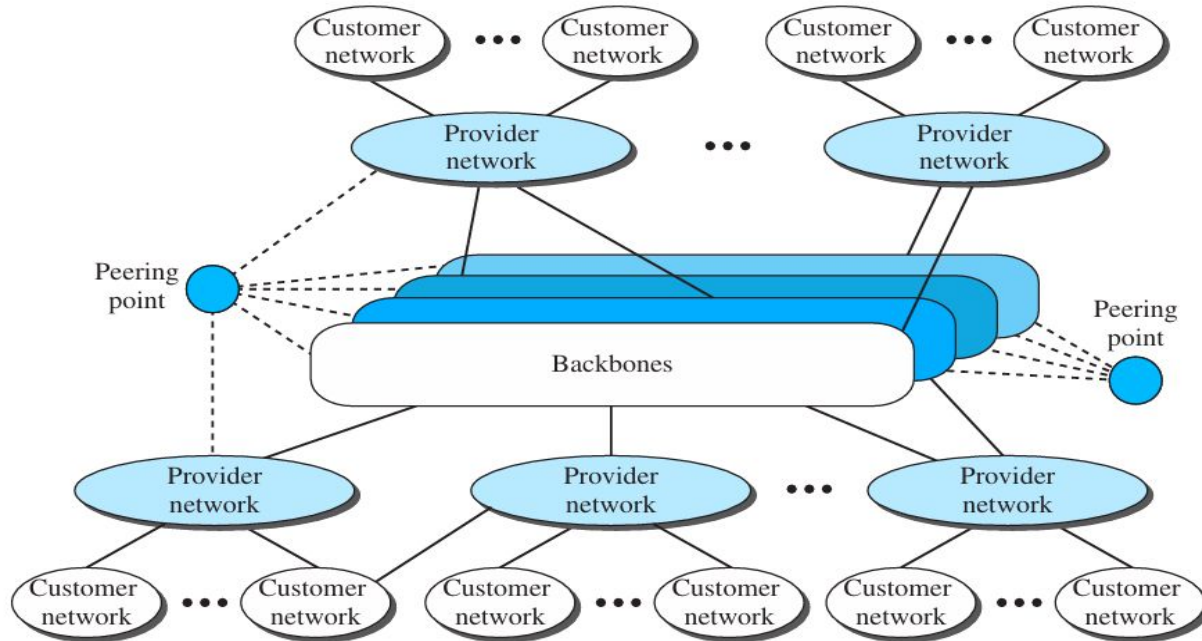
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## 1.2.4. The Internet

- As we discussed before, an internet (note the lowercase i) is two or more networks that can communicate with each other. The most notable internet is called the **Internet** (uppercase I ), and is composed of thousands of interconnected networks. Figure 1.15 shows a conceptual (not geographical) view of the Internet.

**Figure 1.15** *The Internet today*



### 1.2.5. Accessing the Internet

- The Internet today is an internetwork that allows any user to become part of it. The user, however, needs to be physically connected to an ISP. The physical connection is normally done through a point-to-point WAN.

#### Using telephone networks

- Today most residences and small businesses have telephone service, which means they are connected to a telephone network.
- **Dial-up Service** : The first solution is to add to the telephone line a modem that converts data to voice. The software installed on the computer dials the ISP and imitates making a telephone connection.
- **DSL Service** : Since the advent of the Internet, some telephone companies have upgraded their telephone lines to provide higher speed Internet services to residences or small businesses.

#### Using cable Networks

- More and more residents over the last two decades have begun using cable TV services instead of antennas to receive TV broadcasting. The cable companies have been upgrading their cable networks and connecting to the Internet.

#### Using wireless Networks

- Wireless connectivity has recently become increasingly popular. A household or a small business can use a combination of wireless and wired connections to access the Internet.

#### Direct Connection to the Internet

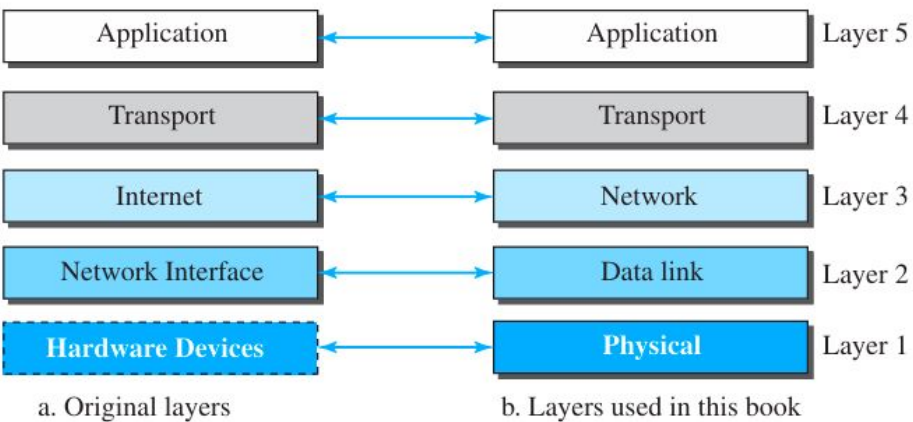
- A large organization or a large corporation can itself become a local ISP and be connected to the Internet.

# 1.3. Network Models:

## 1.3(a). TCP/IP Protocol Suite

- The original TCP/IP protocol suite was defined as four software layers built upon the hardware. Today, however, TCP/IP is thought of as a five-layer model. Figure 2.4 shows both configurations.

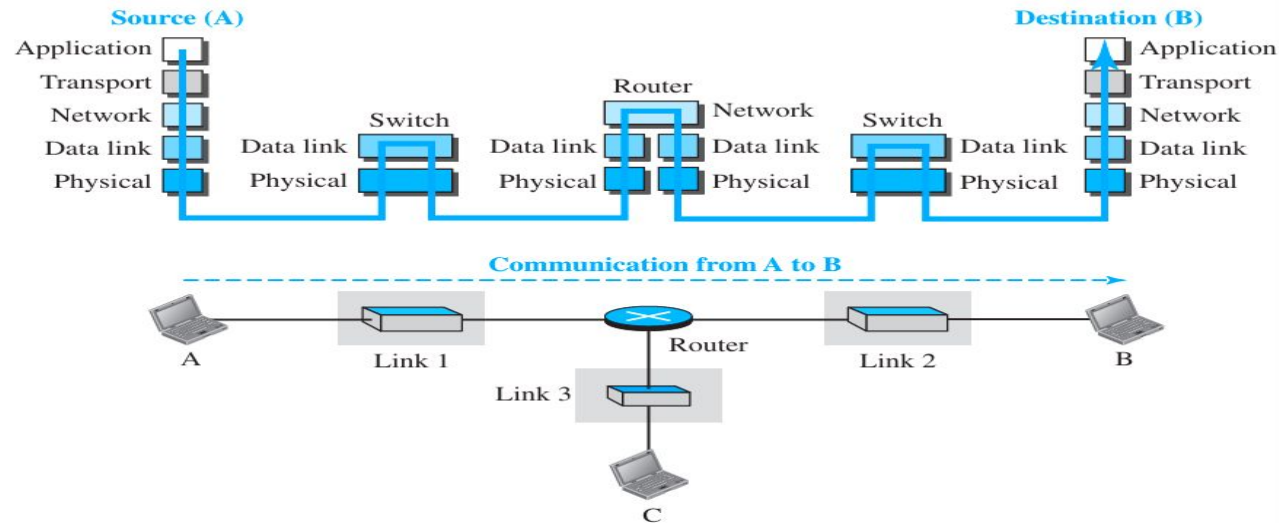
**Figure 2.4** *Layers in the TCP/IP protocol suite*



### 1.3.1(a). Layered Architecture

- To show how the layers in the TCP/IP protocol suite are involved in communication between two hosts, we assume that we want to use the suite in a small internet made up of three LANs (links), each with a link-layer switch. We also assume that the links are connected by one router, as shown in Figure 2.5.
- Let us assume that computer A communicates with computer B. As the figure shows, we have five communicating devices in this communication: (1)**source host (computer A)**, (2)**the link-layer switch in link 1**, (3)**the router**, (4)**the link-layer switch in link 2**, and the (5)**destination host (computer B)**.

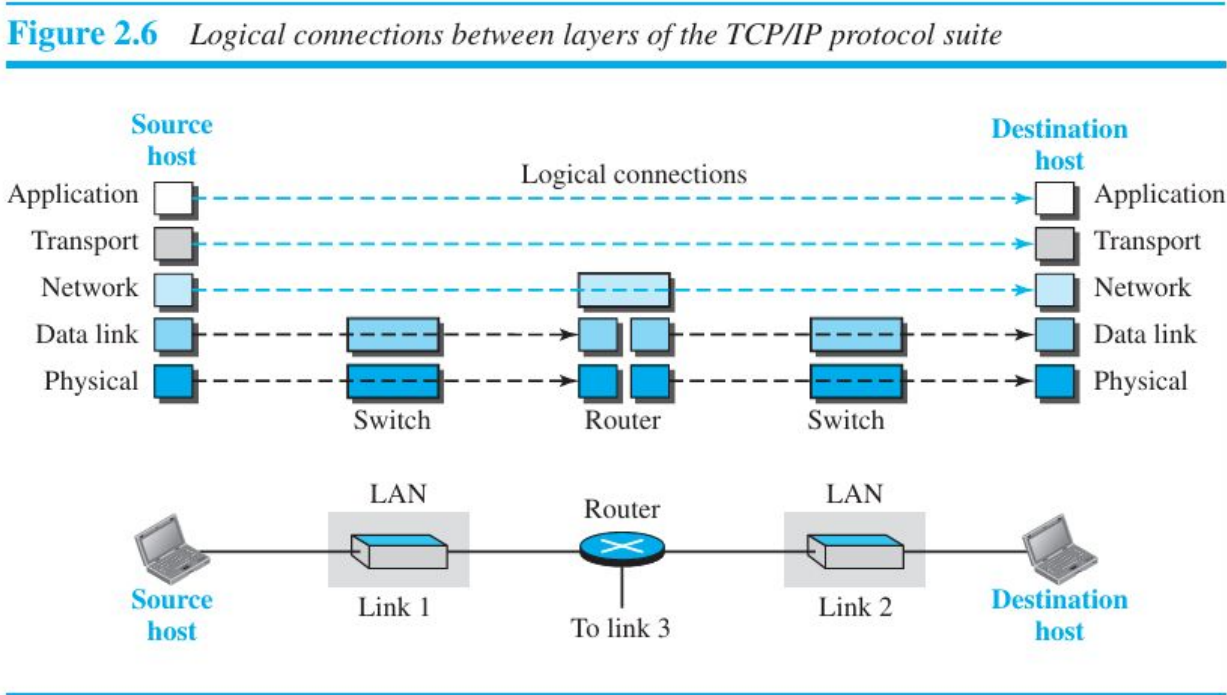
**Figure 2.5** *Communication through an internet*



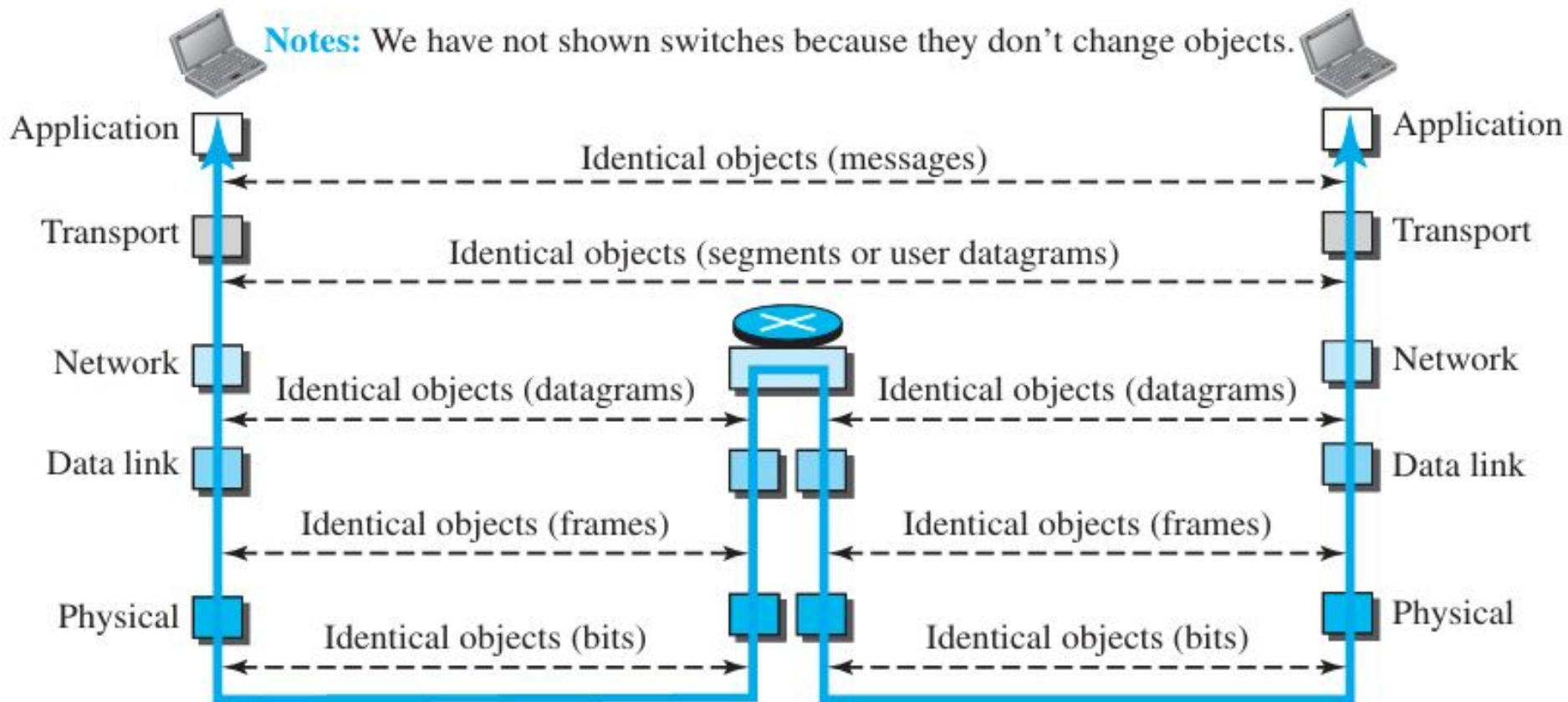


### 1.3.2(a). Layers in TCP/IP Protocol Suit

- To better understand the duties of each layer, we need to think about the logical connections between layers. Figure 2.6 shows logical connections in our simple internet.



**Figure 2.7** *Identical objects in the TCP/IP protocol suite*



### 1.3.3(a). Description of each Layer

#### A) Physical Layer → no specific protocol

- We can say that the physical layer is responsible for carrying individual bits in a frame across the link.

#### B) Data Link Layer → no specific protocol

- We have seen that an internet is made up of several links (LANs and WANs) connected by **routers**. There may be several overlapping sets of links that a datagram can travel from the **host** to the **destination**. The routers are responsible for choosing the best links.

#### C) Network layer → One main protocol(IP), auxiliary protocols(ICMP, IGMP, DHCP, ARP)

- The network layer is responsible for creating a connection and routing between the source computer and the destination computer. The communication at the network layer is host-to-host.

#### D) Transport Layer → Three Protocols(TCP, UDP, SCTP)

- The logical connection at the transport layer is also end-to-end.
- The transport layer at the source host gets the message from the application layer, encapsulates it in a transport layer packet (called a segment or a user datagram in different protocols) and sends it, through the logical (imaginary) connection, to the transport layer at the destination host.

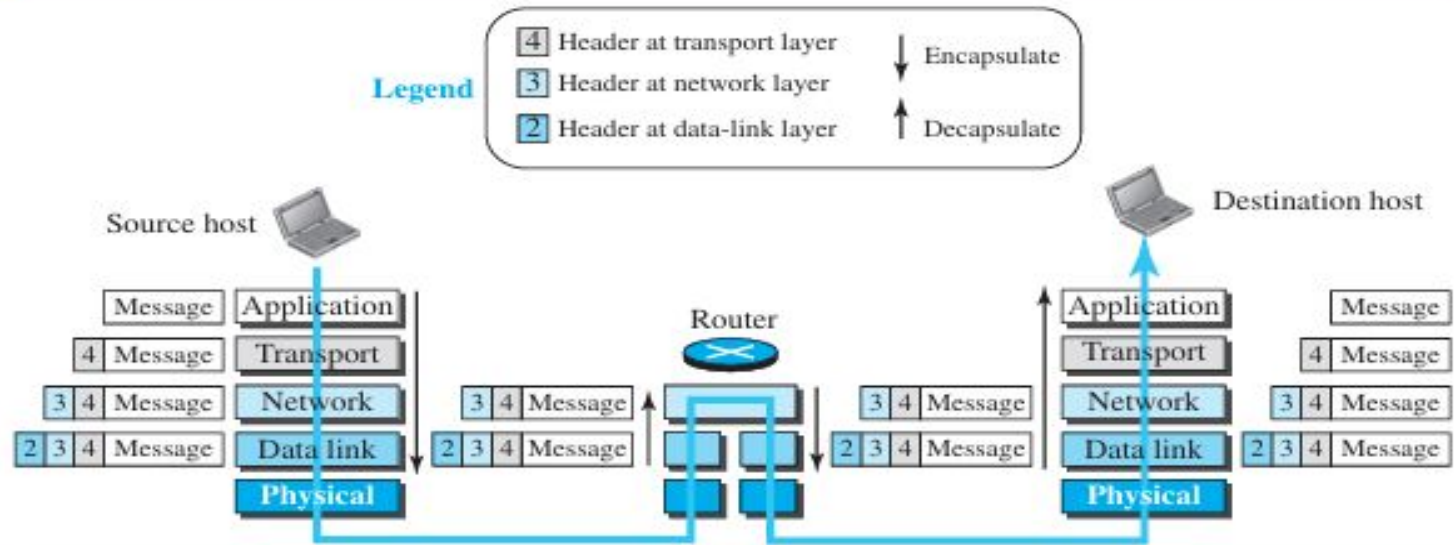
## **E) Application Layer** → HTTP, SMTP, FTP, TELNET, SNMP, DNS, IGMP

- The logical connection between the two application layers is end-to-end. The two application layers exchange messages between each other as though there were a bridge between the two layers.
- **The Hypertext Transfer Protocol (HTTP)** is a vehicle for accessing the World Wide Web (WWW).
- **The Simple Mail Transfer Protocol (SMTP)** is the main protocol used in electronic mail (e-mail) service.
- **The File Transfer Protocol (FTP)** is used for transferring files from one host to another.
- **The Terminal Network (TELNET)** and Secure Shell (SSH) are used for accessing a site remotely.
- **The Simple Network Management Protocol (SNMP)** is used by an administrator to manage the Internet at global and local levels.
- **The Domain Name System (DNS)** is used by other protocols to find the network-layer address of a computer.
- **The Internet Group Management Protocol (IGMP)** is used to collect membership in a group.

### 1.3.4(a). Encapsulation and Decapsulation

- One of the important concepts in protocol layering on the Internet is encapsulation/ decapsulation. Figure 2.8 shows this concept for the small internet in Figure 2.5.

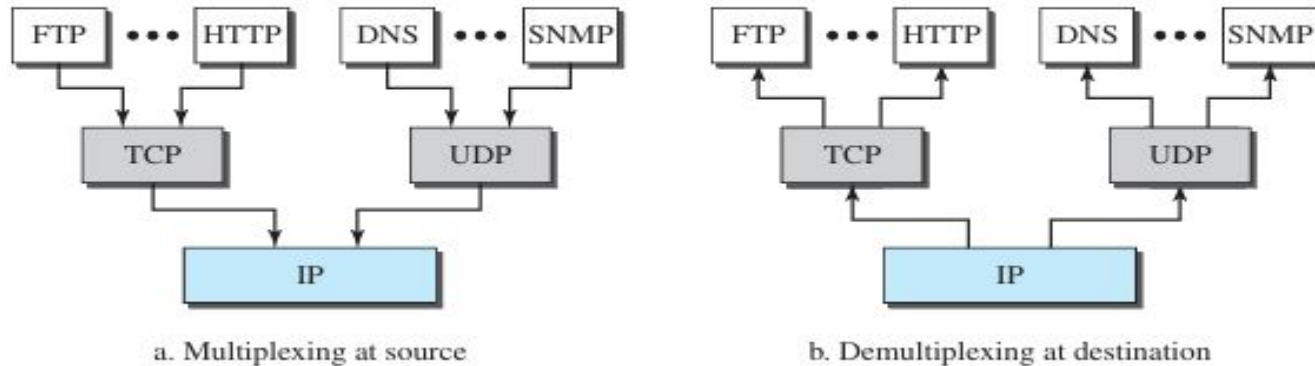
**Figure 2.8**    *Encapsulation/Decapsulation*



### 1.3.5(a). Multiplexing and Demultiplexing

- Since the TCP/IP protocol suite uses several protocols at some layers, we can say that we have multiplexing at the source and demultiplexing at the destination.
- **Multiplexing** in this case means that a protocol at a layer can encapsulate a packet from several next-higher layer protocols (one at a time);
- **Demultiplexing** means that a protocol can decapsulate and deliver a packet to several next-higher layer protocols (one at a time)
- Figure 2.10 shows the concept of multiplexing and demultiplexing at the three upper layers.

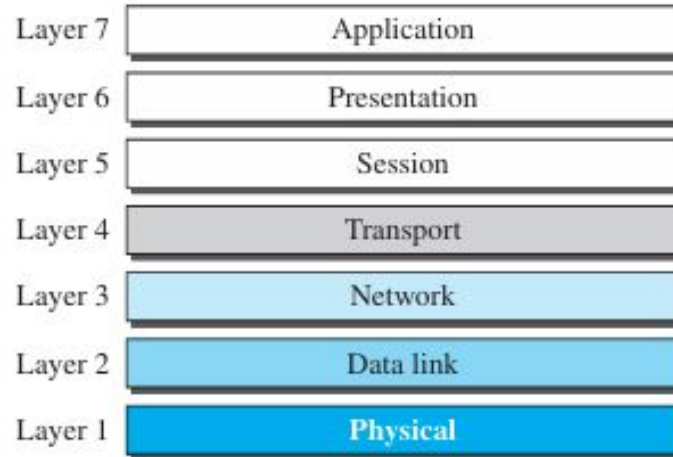
**Figure 2.10** *Multiplexing and demultiplexing*



### 1.3(b). The OSI Model.

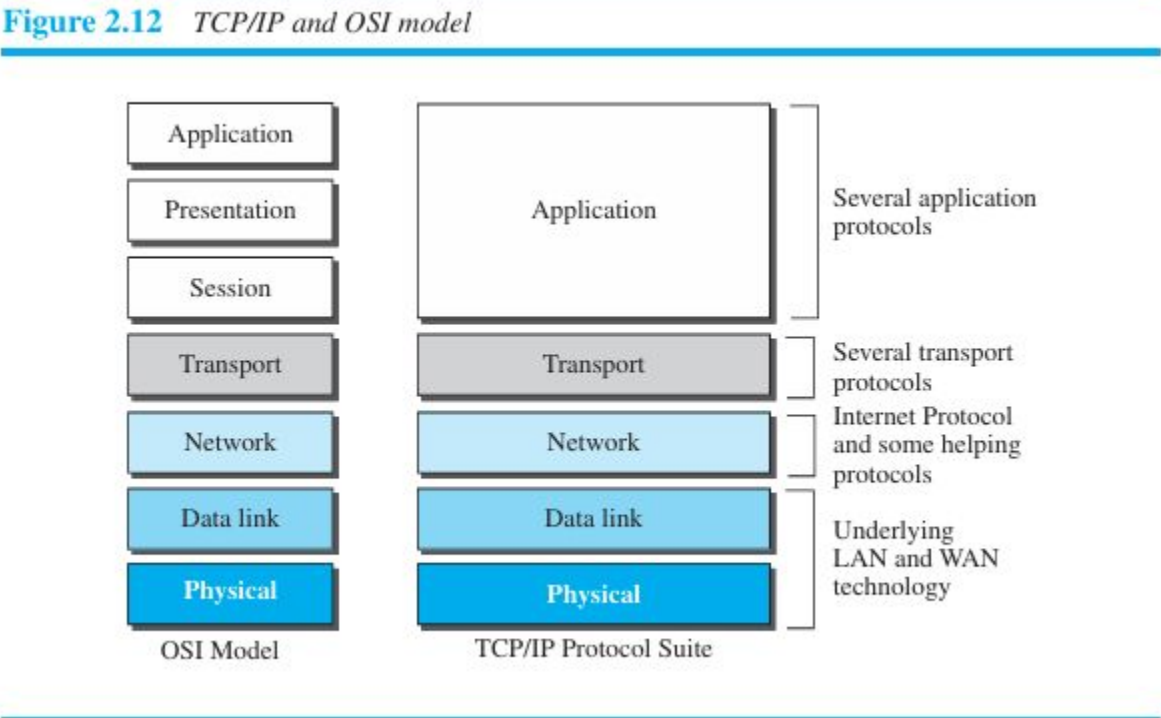
- Established in 1947, the **International Organization for Standardization (ISO)** is a multinational body dedicated to worldwide agreement on international standards. Almost three-fourths of the countries in the world are represented in the ISO. An ISO standard that covers all aspects of network communications is the **Open Systems Interconnection (OSI) model**. It was first introduced in the late 1970s.
- ISO is the organization; OSI is the model.**

**Figure 2.11** *The OSI model*



### 1.3.1(b). OSI vs TCP/IP

- When we compare the two models, we find that two layers, session and presentation, are missing from the TCP/IP protocol suite.
- These two layers were not added to the TCP/IP protocol suite after the publication of the OSI model.
- The application layer in the suite is usually considered to be the combination of three layers in the OSI model, as shown in Figure 2.12.



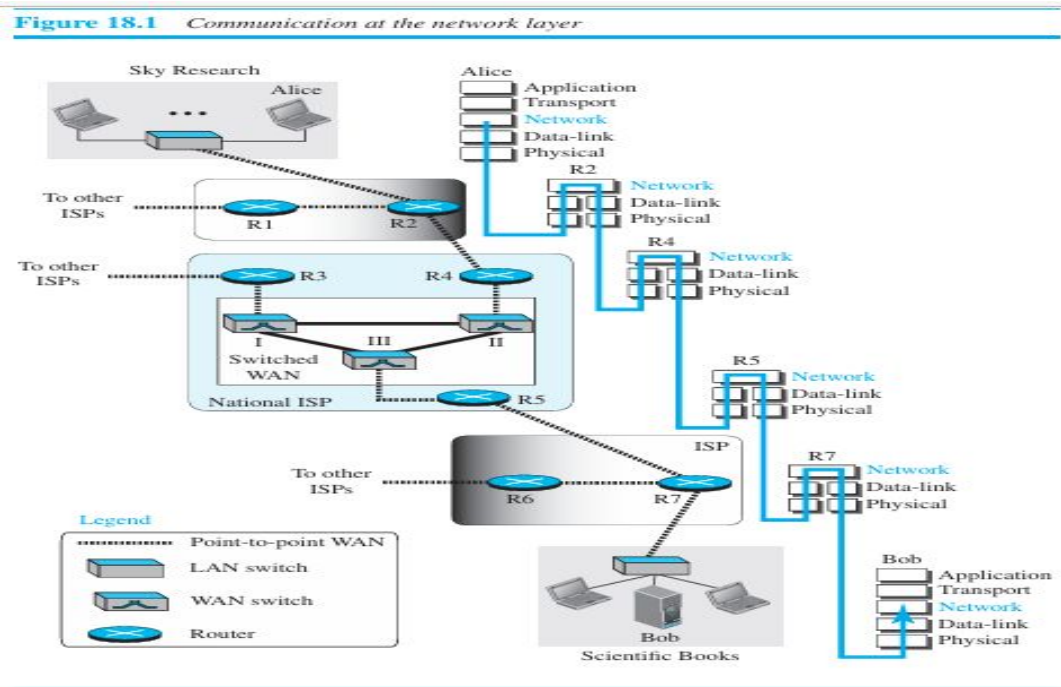


### 1.3.2(b). Lack of OSI Model's Success

- The OSI model appeared after the TCP/IP protocol suite. Most experts were at first excited and thought that the TCP/IP protocol would be fully replaced by the OSI model.

### 1.4. Network Layer: Network layer services

- Figure 18.1 shows the communication between Alice and Bob at the network layer. This is the same scenario we used in Chapters 3 and 9 to show the communication at the physical and the data-link layers, respectively.



### **1.4.1. Packetizing**

- The first duty of the network layer is definitely packetizing: encapsulating the payload (data received from the upper layer) in a network-layer packet at the source and decapsulating the payload from the network-layer packet at the destination.

### **1.4.2. Routing and forwarding**

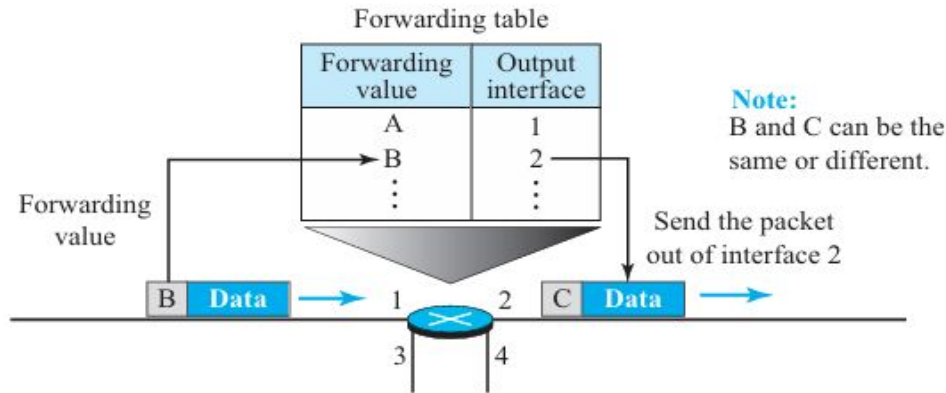
#### **Routing**

- The network layer is responsible for routing the packet from its source to the destination.
- A physical network is a combination of networks (LANs and WANs) and routers that connect them. This means that there is more than one route from the source to the destination.
- The network layer is responsible for finding the best one among these possible routes. The network layer needs to have some specific strategies for defining the best route.

#### **Forwarding**

- If routing is applying strategies and running some routing protocols to create the decision-making tables for each router, forwarding can be defined as the action applied by each router when a packet arrives at one of its interfaces.
- The decision-making table a router normally uses for applying this action is sometimes called the forwarding table and sometimes the routing table.

**Figure 18.2** Forwarding process



### 1.4.3. Other services

#### Error Control

- The designers of the network layer in the Internet ignored this issue for the data being carried by the network layer.
- One reason for this decision is the fact that the packet in the network layer may be fragmented at each router, which makes error checking at this layer inefficient.
- The designers of the network layer, however, have added a checksum field to the datagram to control any corruption in the header, but not in the whole datagram. This checksum may prevent any changes or corruptions in the header of the datagram.

## **Flow Control**

- Flow control regulates the amount of data a source can send without overwhelming the receiver.
- If the upper layer at the source computer produces data faster than the upper layer at the destination computer can consume it, the receiver will be overwhelmed with data.

## **Congestion Control**

- Another issue in a network-layer protocol is congestion control. Congestion in the network layer is a situation in which too many datagrams are present in an area of the Internet.
- Congestion may occur if the number of datagrams sent by source computers is beyond the capacity of the network or routers.

## **Quality of Service**

- As the Internet has allowed new applications such as multimedia communication (in particular real-time communication of audio and video), the quality of service (QoS) of the communication has become more and more important. The Internet has thrived by providing better quality of service to support these applications. However, to keep the network layer untouched, these provisions are mostly implemented in the upper layer.

## Security

- Another issue related to communication at the network layer is security. Security was not a concern when the Internet was originally designed because it was used by a small number of users at universities for research activities; other people had no access to the Internet. The network layer was designed with no security provision. Today, however, security is a big concern. To provide security for a connectionless network layer, we need to have another virtual level that changes the connectionless service to a connection-oriented service.

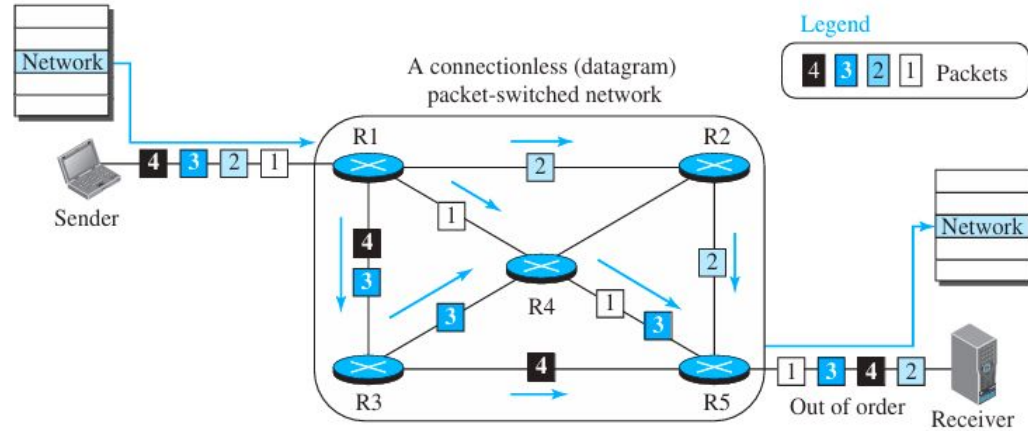
## 1.5 Packet switching

- From the discussion of routing and forwarding, we infer that a kind of switching occurs at the network layer.
- A router, in fact, is a switch that creates a connection between an input port and an output port (or a set of output ports), just as an electrical switch connects the input to the output to let electricity flow.
- Although, in data communication switching techniques are divided into two broad categories, circuit switching and packet switching, only packet switching is used at the network layer because the **unit of data** at this layer is a **packet**.

### 1.5.1. Datagram approach: Connectionless Service

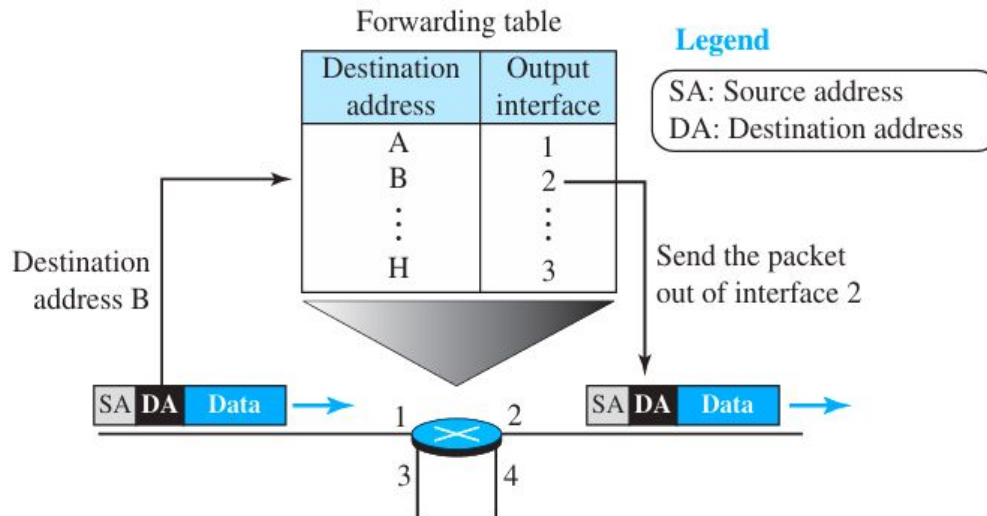
- The network layer was designed to provide a connectionless service in which the network-layer protocol treats each packet independently, with each packet having no relationship to any other packet.
- The idea was that the network layer is only responsible for delivery of packets from the source to the destination.

**Figure 18.3** *A connectionless packet-switched network*



- In the datagram approach, the forwarding decision is based on the destination address of the packet

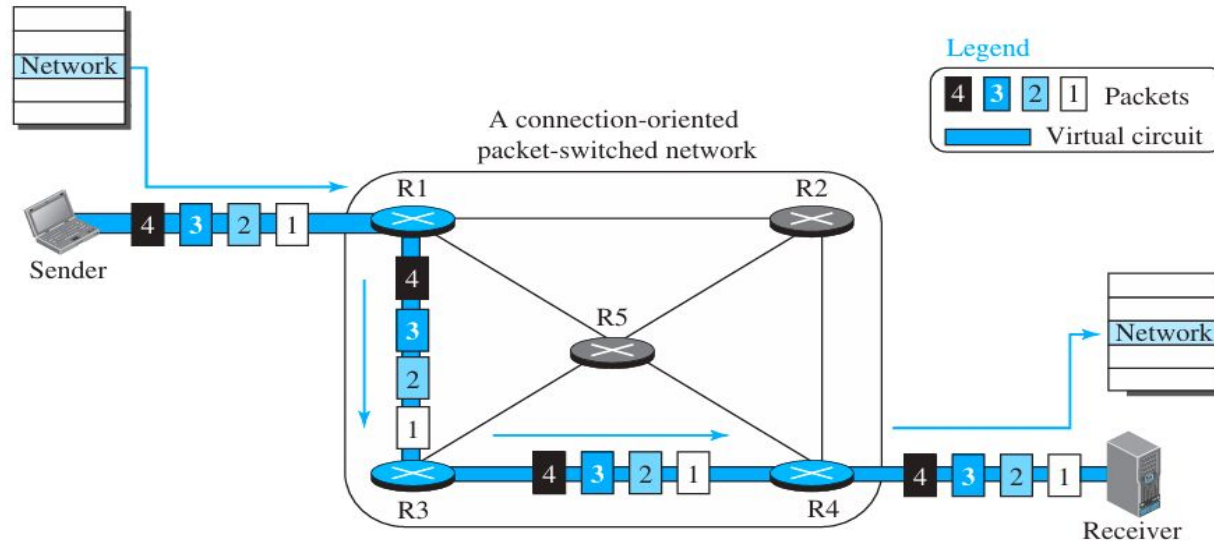
**Figure 18.4** Forwarding process in a router when used in a connectionless network



## 1.5.2. Virtual-Circuit approach

- In the datagram approach, the forwarding decision is based on the destination address of the packet.
- Before all datagrams in a message can be sent, a virtual connection should be set up to define the path for the datagrams.
- After connection setup, the datagrams can all follow the same path.
- In this type of service, not only must the packet contain the source and destination addresses, it must also contain a **flow label** - a virtual circuit identifier that defines the virtual path the packet should follow.

**Figure 18.5** A virtual-circuit packet-switched network





- To create a connection-oriented service, a three-phase process is used: **setup, data transfer, and teardown.**
- In the **setup phase**, the source and destination addresses of the sender and receiver are used to make table entries for the connection-oriented service.
- In the **teardown phase**, the source and destination inform the router to delete the corresponding entries.
- **Data transfer** occurs between these two phases.

## 1.6 Network layer performance

- The upper-layer protocols that use the service of the network layer expect to receive an ideal service, but the network layer is not perfect. The performance of a network can be measured in terms of **delay, throughput, and packet loss.** **Congestion control** is an issue that can improve the performance.

### 1.6.1. Delay

- All of us expect instantaneous response from a network, but a packet, from its source to its destination, encounters delays. The delays in a network can be divided into four types: **transmission delay, propagation delay, processing delay, and queuing delay.**

### 1.6.2. Throughput

- Throughput at any point in a network is defined as **the number of bits passing through the point in a second**, which is actually the transmission rate of data at that point.

### 1.6.3. Packet Loss

- Another issue that severely affects the performance of communication is the number of packets lost during transmission.
- When a router receives a packet while processing another packet, the received packet needs to be stored in the input buffer waiting for its turn.
- A router, however, has an input buffer with a limited size. A time may come when the buffer is full and the next packet needs to be dropped.

### 1.6.4. Congestion Control

- Congestion control is a mechanism for improving performance.
- When the load is much less than the capacity of the network, the delay is at a minimum. This minimum delay is composed of propagation delay and processing delay, both of which are negligible. However, when the load reaches the network capacity, the delay increases sharply because we now need to add the queuing delay to the total delay. Note that the delay becomes infinite when the load is greater than the capacity.
- When the load is below the capacity of the network, the throughput increases proportionally with the load. We expect the throughput to remain constant after the load reaches the capacity, but instead the throughput declines sharply. The reason is the discarding of packets by the routers.
- When the load exceeds the capacity, the queues become full and the routers have to discard some packets.
- In general, we can divide congestion control mechanisms into two broad categories: **open-loop congestion control (prevention)** and **closed-loop congestion control (removal)**.